

Sr.	Unit No.	Question	BL	CO
1	1	Define the following terms. (i) Data (ii) Information (iii) Database	R	C01
2	1	Define the following terms. (i) Field (ii) Record (iii) File	R	C01
3	1	Define the following terms. (i) DBMS (ii) Metadata (iii) Data Warehouse	R	C01
4	1	Explain types of users in database.	U	C01
5	1	List out and explain Application of DBMS	U	C01
6	1	Compare DA Vs DBA.	U	C01
7	1	List out Functions and responsibilities of DBA.	R	C01
8	1	Describe Function and Responsibilities / tasks / roles / duties of DBA.	U	C01
9	1	List out advantages of DBMS.	R	C01
10	1	Explain advantages of DBMS over File Oriented System.	U	C01
11	1	Explain three – level ANSI SPARC database architecture.	U	C01
12	2	Define the terms Schema, Sub-schema and Instance.	R	C02
13	2	Define mapping. Explain internal mapping and external mapping.	U	C02
14	2	Discuss Physical Data Independence and Logical Data Independence.	U	C02
15	2	Draw the structure of Database System Architecture.	R	C02
16	2	Explain execution steps of Database Management System.	U	C02
17	2	Describe functions and services of Database Management System.	U	C02
18	2	List out different types of data model.	R	C02
19	2	Explain Hierarchical and Network Data Model.	U	C02
20	2	Describe Relational Data Model.	U	C02
21	2	Explain Entity – Relational Data Model.	U	C02
22	2	Describe Object Oriented Data Model.	U	C02
23	2	Explain Centralized Database System.	U	C02
24	2	Discuss Parallel Database System.	U	C02
25	2	Write a short note on Client-Server Database System.	R	C02
26	2	Explain Distributed Database System.	U	C02
27	3	Discuss components of SQL. Or Describe DDL, DML, DQL, DCL and TCL.	U	C03
28	3	Describe data types of SQL Server.	U	C03
29	3	Explain CREATE TABLE command with syntax and example.	U	C03
30	3	Illustrate UPDATE command with syntax and example.	U	C03
31	3	State the difference between DELETE and TRUNCATE command.	R	C03
32	3	Write syntax of ALTER command [To add new column and drop existing column].	R	C03
33	3	Illustrate GROUP BY and HAVING Clause with example.	U	C03
34	3	Describe ORDER BY Clause with suitable example.	U	C03
35	3	Explain logical operators in SQL Server. Or Explain AND, OR, NOT Operators.	U	C03
36	3	Describe IN and BETWEEN Operators in SQL Server.	U	C03
37	3	Illustrate LIKE Operator in SQL Server with example.	U	C03

38	3	Explain following Aggregate functions: (i) SUM (ii) AVG (iii) COUNT	U	C03
39	3	Describe the following SQL functions: (i) CEILING (ii) FLOOR (iii) POWER	U	C03
40	3	Explain following functions with example: (i) LEN (ii) SUBSTR (iii) CAST	U	C03
41	3	Describe the following SQL functions: (i) DATEADD (ii) DATEDIFF (iii) EOMONTH	U	C03
42	3	Consider following table and solve given queries using SQL : EMP (emp_no, emp_name, desg, salary, deptno) 1. Create table EMP. 2. Display information of all employees whose name start with 'a' and ends with 'n'. 3. Display department wise salary total. 4. Add new column EmailId in EMP table.	A	C03
43	3	Consider following table and solve given queries using SQL : Employee (eid, ename, department, salary, joining_date, city) 1. Display the highest, lowest, total and average salary of all employees. 2. List the departments having total salaries more than 35000 and located in 'Surat' 3. Display the total number of employees hired after 31 st December, 2001.	A	C03
44	3	Consider following table and solve given queries using SQL : Customer (cno, cname, address, city, contact_no) 1. Display all rows and columns of a Customer table. 2. Display all the information of Customer table according to city name in descending order. 3. Display only customer id and customer name who belongs to 'ahemdabad' city. 4. Destroy Customer table from the database.	A	C03
45	3	Consider following tables and solve given queries using SQL : Sailors (sid, sname, rating, age) Boats (bid, sid, color) 1. Display all the details of Sailors and Boats table (Use appropriate Join operation). 2. Find the name and age of youngest sailor. 3. Find the sailor names who sails 'red' color boat.	A	C03
46	3	Consider following tables and solve given queries using SQL : Customer (customer_id, customer_name, city, salesman_id) Salesman (salesman_id, salesman_name, commission) Display customer details who live in 'New York', 'Paris', 'London'. (Use In). 1. Write a query to display first three customer details. 2. Write a query to fetch all the details of customer whose name is not started from letter 'P'. 3. Change the name of customer 'Aric' to 'Mr. Aric Brook'. 4. Add one more column contact_number int not null to customer table. 5. Delete all the details of salesman table whose salesman id is between 218 and 233. 6. Perform left outer join between Customer and Salesman table and display customer name, salesman name and commission columns.	A	C03
47	4	Explain following Relational Algebra Operation with example. 1. Selection Projection	R	C04
48	4	Explain following Relational Algebra Operation with example. Cross Product	R	C04
49	4	Explain following Relational Algebra Operation with example.	U	C04

		1. Joins (Inner Join, Outer Joins) 2. Rename Division		
50	4	Explain Set Operators in Relational Algebra. Or Explain Union, Intersection, Minus in Relational Algebra.	U	C04
51	4	Consider following table and solve given queries using relational algebra . Customer (cid, cname, address, city, contact_no) 1. Display name and city of customer whose name is 'Anil' and belongs to 'Goa' city. 2. Display the details of Customers having customer id between 101 and 105. 3. Display only customer id and customer name who belongs to 'Ahemdabad' city. 4. Display customer id, customer name and city of customer having customer id is 103 and not belongs to 'Baroda' city.	A	C04
52	4	Consider following table and solve given queries using relational algebra . Student (Rollno, Name, Branch) Result (Rollno, SPI) 1. Combine information of Student and Result table using cross join. 2. Display Rollno, Name, Branch and SPI of EC branch students only. 3. Display Rollno and Name of the student along with their SPI.	A	C04
53	4	Consider the following relation and write a relational algebra expression . Car (Car_ID, Car_Name, Description) 1. Display the car name along with the description. 2. Display the car id and name from Car relation where car name is "Tata Indica". 3. Display the car name and description where car id is 103 and car name is "Mercedes-Benz".	A	C04
54	5	Define the following terms: (i) Entity (ii) Attribute (iii) Relationship	R	C05
55	5	Explain E-R diagram with its symbols.	U	C05
56	5	Discuss different types of Attributes in E-R Model.	U	C05
57	5	Describe Mapping Cardinality in E-R Model.	U	C05
58	5	Define Weak Entity, Strong Entity and Weak Relationship with suitable example.	R	C05
59	5	Describe Participation constraints with suitable example. Or Describe Total and Partial participation with suitable example.	U	C05
60	5	Discuss Specialization with suitable diagram	U	C05
61	5	Explain Generalization with diagram.	U	C05
62	5	Explain NOT NULL and UNIQUE Constraint with suitable example.	U	C05
63	5	Describe CHECK Constraint with suitable example.	U	C05
64	5	Explain PRIMARY Key and FOREIGN Key Constraint with suitable example.	U	C05
65	5	State the difference between PRIMARY Key and UNIQUE Key.	R	C05
66	5	Prepare E-R diagram of Bank Management System that include Specialization and Generalization symbols also.	A	C05
67	5	Prepare E-R diagram of the following Systems and assume suitable entities and attributes. 1. Hospital Management System 2. Library Management System 3. College Management System	A	C05